

Suzzana Rafi

Curriculum Vitae

Education

Fall 2021-Present — **PhD in Computer Science**, *George Mason University*, USA, *Current GPA – **3.92** (On a scale of 4.0)*.
Advisor: Wing Lam, Expected Graduation: May 2027

Fall 2021-Spring 2024 — **MSc. in Computer Science**, *George Mason University*, USA, *Current GPA – **3.87** (On a scale of 4.0)*.

2014-2018 — **B.Sc. in Computer Science and Engineering**, *Bangladesh University of Professionals (BUP)*, Bangladesh.

Objective

My current research is focused on Software Testing and Analysis, particularly on enhancing testing techniques through the use of AI techniques. I have experience working on detecting, repairing and reproducing flaky tests, which are tests that pass and fail non-deterministically on the same version of code, and have built heuristics and large language model based tools to address testing problems.

Computer skills

Programming languages	Bash, Python, Java, C, C++, HTML, CSS, PHP, SQL, Assembly, JavaScript, OpenGL
DBMS	MySQL, Oracle 11g
Frameworks	Bootstrap, Metasploit
Tools	Maven, JaCoCo, NonDex, iDFlakies, PITest, EclEmma, Checkstyle, Emacs, Tmux, Android Studio, MATLAB, Weka, Wireshark, NMAP, Nessus, John The Ripper, THC Hydra, CISCO Packet Tracer, Autodesk 3ds Max, QuickBooks, HubSpot, Insightly,Adobe Photoshop, Adobe Illustrator

Publications

2026 Suzzana Rafi, Mahbub-Ul-Hoque Sumon, MD Erfan, Maruf Morshed Khan, August Shi, Wing Lam. **A Dataset of Reproducible Flaky-Test Failures** *Proceedings of ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA)*, 2026.

2026 Suzzana Rafi, Md Mahmudul Hasan Pious, Shanto Rahman, August Shi, Wing Lam. **Optimizing Search for Tests Relevant to Order-Dependent Flaky Tests** *Proceedings of the IEEE International Conference on Software Maintenance and Evolution (ICSME)*, 2026.

2025 Shanto Rahman, Bala Naren Chanumolu, Suzzana Rafi, August Shi, Wing Lam. **Ranking Relevant Tests for Order-Dependent Flaky Tests** *IEEE/ACM International Conference on Software Engineering (ICSE)*, 2025.

2023 Sajed Jalil, Suzzana Rafi, Thomas D. LaToza, Kevin Moran, Wing Lam. **ChatGPT and Software Testing Education: Promises & Perils.** *IEEE International Conference on Software Testing, Verification and Validation Workshops (ICSTW)*, 2023.

2020 Muhammad Nazrul Islam, Toki Tahmid Inan, Suzzana Rafi, Syeda Sabrina Akter, Iqbal H Sarker, AKM Najmul Islam. **A Systematic Review on the Use of AI and ML for Fighting the COVID-19 Pandemic.** *IEEE Transactions on Artificial Intelligence*, 2020.

2020 Mosaddek Hossain, Suzzana Rafi, Shohrab Hossain. **An Optimized Decision Tree-based Android Malware Detection Approach Using Machine Learning.** *International Conference on Networking, Systems and Security (NSysS)*, 2020.

2019 Sadia Akter, Tania Tahmina Jui, Tasbiraha Athaya, Afsana Zaman, Suzzana Rafi. **A Proposed System for Fare Measurement for Rickshaws in Bangladesh.** *International Conference on Advances in Electrical Engineering (ICAEE)*, 2019.

2018 Shusmoy Kundu, Khandaker Annatoma Islam, Tania Tahmina Jui, Suzzana Rafi. **Cyber Crime Trends in Bangladesh: An Analysis and Ways to Combat the Threat.** *International Conference on Advanced Communications Technology (ICACT)*, 2018.

Research Experiences

1. **A Dataset of Reproducible Flaky-Test Failures**
   Tools: Docker, Maven, JaCoCo, Checkstyle, NonDex, iDFlakies, NIODebugger
   Languages: Bash, Python, Java
   Description: Flaky tests may pass or fail inconsistently when executed on the same code version, making them difficult to reproduce and repair. This work introduces ReproFlake, a dataset of 1,115 reproducible flaky tests. The dataset includes reproducible environments, reproduction and validation scripts, and execution logs to support research on flaky-tests. The work also analyzes the challenges of reproducing flaky failures and the characteristics of flaky-test fixes across different flakiness categories.

2. **Optimizing Search for Tests Relevant to Order-Dependent Flaky Tests**
   Languages: Python, Bash, Maven
   Description: This work improves the process of finding the tests responsible for these failures by enhancing the delta-debugging search used in iFixFlakies. The proposed techniques use hierarchical information, historical execution data, TF-IDF similarity, and LLM-based similarity to guide the search and reduce the number of test executions needed to locate relevant tests.

2. **Ranking Relevant Tests for Order-Dependent Flaky Tests**
   Tools: Large Language Models, srcML, Maven
   Languages: Bash, Python
   Description: Regression testing faces challenges from flaky tests, which are tests that may pass or fail inconsistently for the same code version. A substantial type of flaky test is the order-dependent flaky test, where the outcome depends on the test execution order. This work introduces two methods to rank tests that causes order-dependent failures. The first method uses LLM to analyze the content of test methods, and the second method employs statistical analysis on past test-order execution data.

2. **ChatGPT and Software Testing Education: Promises & Perils**
   Tool: ChatGPT
   Languages: Bash, Python
   Description: We examine how well ChatGPT performs when tasked with answering common questions in a popular software testing curriculum.

4. **A Systematic Review on the Use of AI and ML for Fighting the COVID-19 Pandemic**
   Description: Performed a systematic review of AI/ML approaches for COVID-19, analyzing 49 selected studies to synthesize models, data types, applications, and key research gaps.

5. **Android Malware Detection using Machine Learning**
   Tool: Weka
   Languages: Python
   Description: A project on Android malware detection using machine learning techniques. Our project shows that decision tree based algorithms perform best in identifying malware from benign applications.

7. **Cyber Security – Study and Analysis of Threats and Vulnerabilities & Hardening of Operating System**
   Framework: Metasploit
   Tools: NMAP, Nessus
   Description: A study of cyber security fundamentals, practical implementation of gaining remote access within a network. We also hardened an OS (Windows 10) and were able to make a comparison of pen-testing the hardened and normal OS.

Research Interests

Software Testing, Software Engineering, Bug Detection and Repair, Large Language Models, Artificial Intelligence, Machine Learning, Deep Learning, Data Mining, Human Computer Interaction, Cybersecurity, Network Security, Cryptography, User Experience

Achievements and Participations

2026 Awarded financial support from College of Engineering and Computing (CEC) Software Engineering Scholarship Fund

2024 Received Outstanding Academic Achievement Award

2023 Received Outstanding Graduate Teaching Assistant award

Summer 2022-2026 Mentored Aspiring Scientists Summer Internship Program (ASSIP) students at George Mason University

2024, 2023 Participated in CRA-WP Grad Cohort for Women

2024 Participated in Grace Hopper Celebration

2024 Participated in ESSEC - Eastern Atlantic Students in Software Engineering Colloquium

2023 Participated in ISSTA Doctoral Symposium 2023

2017 Participated in a competition "International Humanitarian Technology Project Competition (IHTPC)" for "**Smart Rickshaw**: An Automated Fare System"

2017 Participated in a competition "Digital Innovation Challenge for Women", Microsoft, 2017 for "**Smart Rickshaw**: An Automated Fare System"

2011, 2013 Achieved board scholarship for excellence in both Secondary School Certificate & Higher Secondary Certificate examination

Work Experiences

Fall 2021-Present Graduate Research Assistant, Graduate Teaching Assistant, George Mason University

2019 Digital Transformation Analyst, Humans & Software, Bangladesh

2016-2017 Industrial Trainee, Grameenphone Limited, Bangladesh

Academic Projects

1. **Emotion Detection from Text: Opinion Mining using Machine Learning and Deep Learning Approaches**
   Languages: Python
   Description: A project on sentiment and emotion analysis which are key methods to identify and measure feelings. We detect emotions from texts via Machine/Deep learning and NLP (Natural Language Processing).

2. **ANCA Tracker(Mobile Application & Web based Software)**
   Languages: Java, HTML, CSS, JavaScript, PHP, SQL
   Description: A mobile application and web based software about health care which takes daily calorie intake and calculates how to reach target goal. Health tips and medical help are also available.
3. **Smart Menu System(Software & Hardware Project)**
   Languages: Java, C++, SQL
   Description: A smart menu system which recommends food according to time, weather and popularity. Menu is displayed on a graphical UI using Java Swing. Weather is measured using Arduino. The menu and temperature were saved in the database.
4. **Database Management System of Leather Industry**
   Languages: HTML, CSS, PHP, SQL
   Description: The database management system for every sector of leather industry was created. We used Oracle 11g for maintaining the database.
5. **Programming From Scratch(Web Based Software)**
   Framework: Bootstrap
   Languages: HTML, CSS, JavaScript, PHP, SQL
   Description:A programming learning site with tutorials for beginners.
6. **Smart Rickshaw: An Arduino Based Automated Fare System (Hardware Project)**
   Languages: C++
   Description: A micro-controller based fare system which calculates rickshaw fare according to distance, slope, weather and number of passengers.